P.5 Exponential and Logarithmic Functions

Name _____ Period ____ Date _____

AP Calculus P.5 · Student sheet

I can…

- graph exponential functions and give their domain, range, and asymptote.
- explain what the number e is and use natural logarithms.
- find the domain of a logarithmic function and solve exponential and logarithmic equations.

Vocabulary · fill in the missing word

Exponential function — $f(x) =$ _____, where $a > 0$ and $a \neq 1$

The number e — the base of the _____ function whose tangent line at (0, 1) has slope 1; $e \approx 2.718$

Logarithmic function — $y =$ _____ $= a^y$, where $a > 0$, $a \neq 1$, and $x > 0$

Natural logarithm — $\ln x =$ _____

Part 1 · Find e

Graph $y = a^x$ (slider for a) and $y = x + 1$.

a	2	2.5	2.7	3
Does the line cross or touch the curve at (0, 1)?				

The best value of a: _____

Part 2 · Notes and examples (fill in each note, then work the example)

$f(x) = a^x$: domain all real numbers, range $(0, \infty)$, horizontal asymptote $y = 0$, and the graph contains (0, 1), (1, a), and (−1, _____).

Example 1

Use transformations to graph $f(x) = 2^{x+2}$. Give the domain, range, and asymptote.

The number e: the tangent line to $y = e^x$ at (0, 1) has slope _____.

Example 2

Graph $f(x) = 5 - e^x$. Give the range and asymptote.

Logarithms: $y = \log_a x$ means $x =$ _____; the domain of $\log_a x$ is $(0, \infty)$.

Example 3

Find the domain of each function.

a. $F(x) = \log_2 x^2$

b. $g(x) = 8 + 5 \ln(2x + 3)$

c. $f(x) = \ln(x - 1)$

Inverse properties: $\ln e^x = x$ for all real x, and e^(ln x) = _____ for $x > 0$.

Example 4

Solve.

a. $3^{x^2} = 9^x$

b. $e^{3x} = \frac{e^2}{e^x}$

Example 5

Solve.

a. $\log_2(2x + 1) = 3$

b. $\ln(2x + 3) = 2 \ln 3$

Part 3 · Partner practice

Partner 1

Solve $e^{1 - 2x} = 4$. Give the exact answer and a decimal.

Partner 2

Solve $\log_3(3x - 2) = 2$.